

IEEE Recommended Practice for Applying Low-Voltage Circuit Breakers Used in Industrial and Commercial Power Systems

Chapter 1 Overview

1.1 Scope

This recommended practice provides information for selecting the proper circuit breaker for a particular application. This recommended practice helps the application engineer specify the type of circuit breaker, ratings, trip functions, accessories, acceptance tests, and maintenance requirements. It also discusses circuit breakers for special applications, e.g., instantaneous only and switches. In addition, it provides information for applying circuit breakers at different locations in the power system and for protecting specific components. Guidelines are given for coordinating combinations of line-side and load-side devices. Acceptance testing and maintenance guidelines are provided so that reliable operation can be verified and maintained.

This recommended practice does not cover the selection and application of circuit breakers such as marine circuit breakers and definite purpose circuit breakers.

1.2 Two classifications of breakers

There are two main classifications of low-voltage circuit breakers: molded-case circuit breakers and low-voltage power circuit breakers. Within the molded-case circuit breaker classification, there is another type of circuit breaker called the insulated-case circuit breaker. The construction and characteristics of these three types will be discussed. Throughout the balance of this recommended practice, these devices will be referred to as follows:

- MCCB: molded-case circuit breaker
- ICCB: insulated-case circuit breaker
- LVPCB: low-voltage power circuit breaker

Each one of these circuit breakers has different design characteristics and, in many cases, different application requirements.

This recommended practice compares the different types of circuit breakers so that the power systems engineer can decide which one is best suited for a particular application. In addition, it discusses ratings, such as overload, short-time, and interrupting capabilities. Protection requirements depend on the circuit breaker location in the power system as well as the type of equipment that is being protected. Examples for different types of equipment and circuit locations are discussed in this recommended practice.

MCCBs are tested and rated in accordance with UL 489-2002.¹ Their current-carrying parts, mechanisms, and trip devices are completely contained within a molded case of insulating material. The cover and base of smaller MCCBs are designed so that the MCCBs cannot be opened for maintenance purposes. The main contacts of MCCBs cannot be removed; however, some MCCBs are available with field-installable accessories. MCCBs are available in stationary or plug-in construction with circuit-breaker enclosures that can be flush or surface mounted. They are available in a large number of continuous-current and interrupting ratings. The smaller continuous-current ratings are equipped with thermal-magnetic or magnetic only trip units. Larger sizes are also available with thermal-magnetic or electronic (static) trip devices.

ICCBs are also tested and rated in accordance with UL 489-2002. As with MCCBs, ICCB current-carrying parts, mechanisms, and trip units are contained within a molded case of insulating material. The case is designed so that it can be opened for inspection of contacts and arc chutes and for limited maintenance. Most manufacturers offer designs that permit replacement of accessories, and some designs permit replacement of the main contacts. ICCBs are available in both stationary and drawout construction. They are generally characterized by a stored energy mechanism, larger frame sizes, and higher short-time withstand ratings than MCCBs. Electronic trip units are standard.

LVPCBs are tested and rated according to the following standards:

- ANSI C37.16-2000
- ANSI C37.17-1997
- ANSI C37.50-1989 (R2000)
- IEEE Std C37.13TM-1990
- UL 1558-1999

LVPCBs are generally characterized by physically large frame sizes, drawout construction, and the highest short-time withstand ratings of all the types of low-voltage circuit breakers. When the circuit breaker is removed from its enclosure, the current-carrying parts and operating parts are accessible for inspection, maintenance, and replacement purposes. Electromechanical trip units were used in the circuit breakers prior to the early 1970s. However, electronic trip units are used in new LVPCBs and are available as upgrades for older units.

1.3 Description of a molded-case circuit breaker

Figure 1-1 is a cutaway view of a typical MCCB. Letters are used to indicate the various elements of the circuit breaker, with a description listed in the legend. This typical circuit

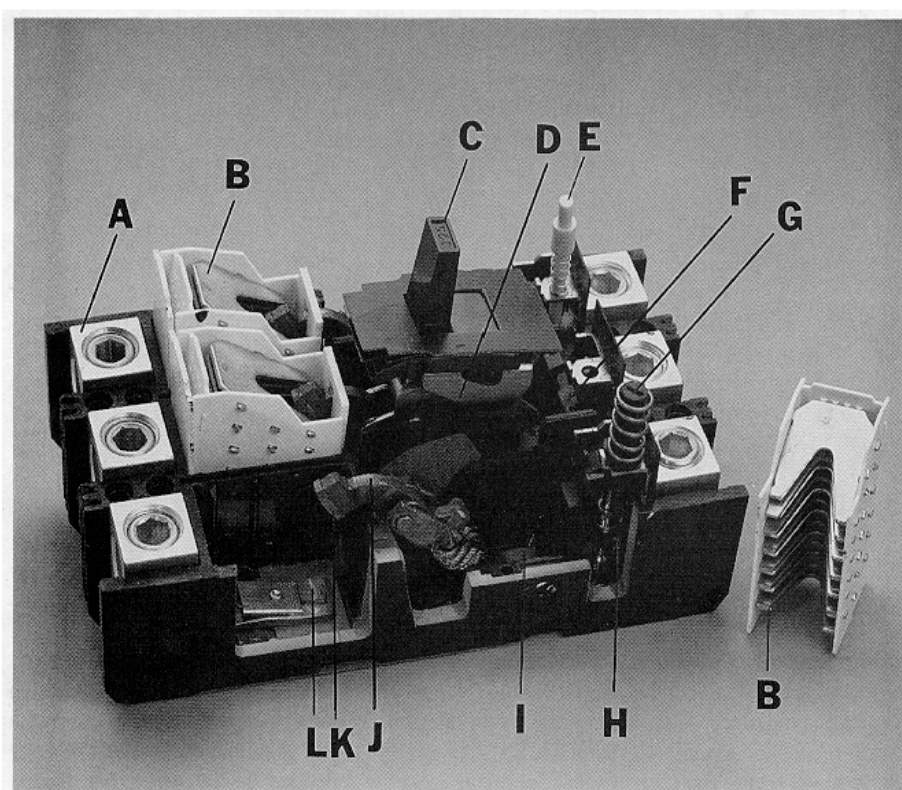
¹Information on references can be found in 1.7.

breaker operates using an over-center toggle, quick-make-quick-break mechanism. This mechanism is operated manually to the ON (closed) and OFF (open) positions using the handle. The quick-make-quick-break action ensures that the speed at which the breaker contacts are open or closed is independent of the speed at which the handle is moved. This toggle mechanism is also trip-free, which means that the circuit breaker cannot be prevented from tripping by holding or locking the handle in the ON position. When the circuit breaker trips open automatically, the handle will assume either an intermediate position between ON and OFF or the OFF position. If the handle moves to the intermediate position, it must be manually moved slightly past the OFF position to reset the mechanism. Instructions for resetting a particular circuit breaker after it trips should be marked on the circuit breaker and/or indicated on the equipment where the circuit breaker is installed.

1.4 Description of a low-voltage power circuit breaker

Figure 1-2 is a view of a partially disassembled, manually operated, drawout LVPCB. The open construction permits access to the circuit-breaker parts for maintenance and parts replacement. Numbers are used to indicate the various elements of the circuit breaker. A description of each element is listed in the legend. The following description of the operation of the circuit breaker starts with the open position. The circuit breaker condition “open” is indicated on the face of the circuit breaker. To close the circuit breaker, a spring mechanism must be charged. The springs are charged by pulling down and releasing the manual spring charging handle. The spring condition “charged” is indicated on the face of the circuit breaker. The circuit breaker is manually closed by depressing the close (push-to-close) hood. The circuit breaker condition “closed” is indicated on the face of the circuit breaker. The circuit breaker is opened manually by depressing the open (push-to-trip) lever or automatically by the operation of the trip unit.

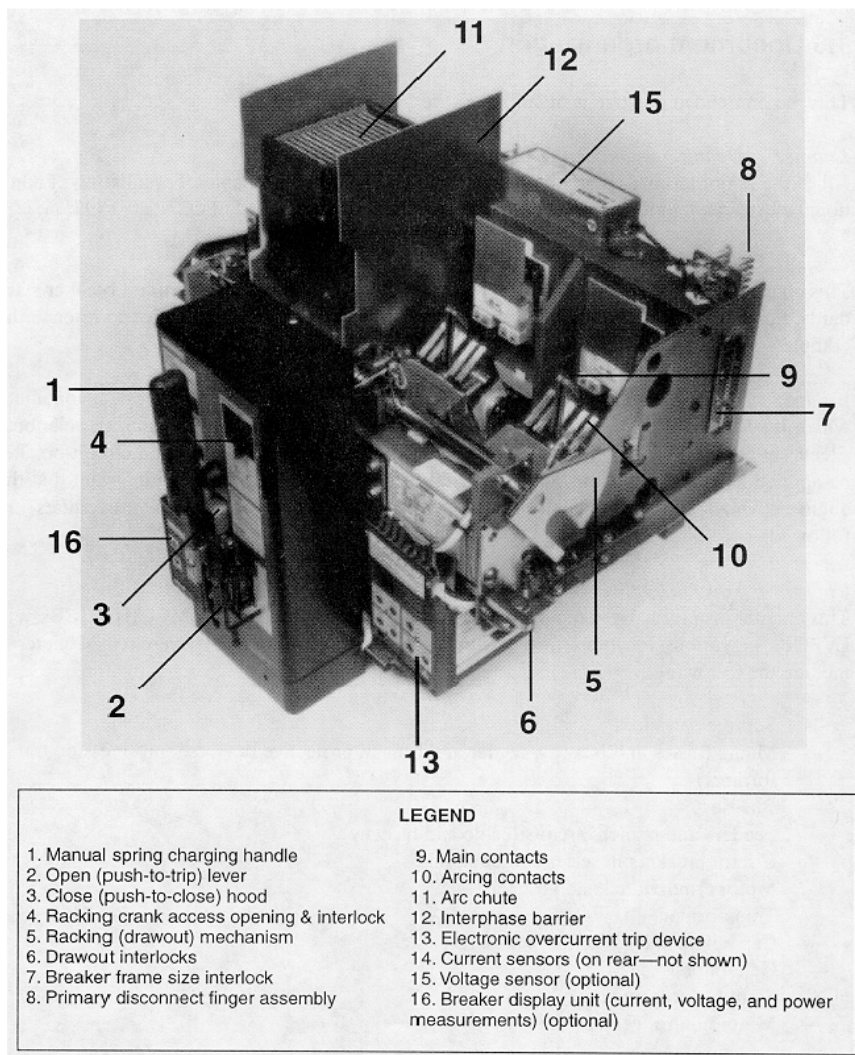
The drawout circuit breaker has three separate positions: “CONNECT,” “TEST,” and “DISCONNECT.” A racking crank is used to move the drawout circuit breaker to each position in the circuit breaker compartment. The circuit breaker's contacts are only connected to the external power circuit in the connected position. If the circuit breaker is closed in the TEST position, there is no effect to the external power circuit. Interlocking prevents moving a closed circuit breaker between these positions or closing it in other than the CONNECT or TEST position. Further interlocking prevents inserting circuit breakers of the wrong frame size into a compartment. Primary disconnects and optional secondary disconnects automatically complete the power circuit in the CONNECT position and control circuits in the CONNECT and TEST positions, respectively.



LEGEND	
A.	Wire connector
B.	De-ionizing arc stack
C.	Handle
D.	Operating mechanism
E.	Test trip actuator
F.	Common-trip bar
G.	Instantaneous trip level adjustment
H.	Electro-magnet
I.	Bimetal
J.	Moving arm
K.	Moving contact
L.	Stationary contact

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Figure 1-1—Cutaway view of a typical MCCB



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**Figure 1-2—Low-voltage ac power circuit braker-drawout type
(shown partially disassembled to show internal features)**

1.5 Document organization

This recommended practice is organized in the following manner:

Chapter 2: Definitions, acronyms, and abbreviations

All of the major terms used in this recommended practice are defined in this chapter. For a listing of additional electrical definitions, refer to *The Authoritative Dictionary of IEEE Standards Terms* [B1]² and IEEE Std C37.100TM-1992.

Chapter 3: Rating and testing

This chapter summarizes the application requirements of recognized low-voltage circuit breaker standards. For more details on a particular application, the engineer is encouraged to refer to the complete standard for amplification and a more complete discussion.

By understanding the differences in standards, an engineer can make a better decision about which type of breaker should be used for a particular application. In addition, Section 110.3(B) from the National Electrical Code[®] (NEC[®]) (NFPA 70-2005) requires that a circuit breaker be applied according to the information listed in the standards. Thus, it is important that the engineer know and understand the standards so that proper application procedures are followed.

Chapter 4: Specific applications

This chapter provides a systematic procedure for selecting and applying MCCBs, ICCBs, and LVPCBs at various locations in a power system. The applications covered in Chapter 4 are as follows:

- Service entrances
- Mains (buses and busway, feeder and branch protective devices, and line-side transformers)
- Bus ties
- Feeders and branch circuits (cable and busway)
- Circuit breakers in series combinations
- Motors (individual and grouped)
- Transformers
- Capacitors and capacitor banks
- Generators
- Switchboards and panelboards
- Motor control centers and starters

²The numbers in brackets correspond to those of the Bibliography in 1.8.

Chapter 5: Selective coordination of low-voltage circuit breakers with other protective devices

This chapter explains how to coordinate different combinations of devices and provides typical settings. Conflicting objectives normally occur between protection and selective coordination. The objective of protection is to minimize the damage by removing the overload or short circuit as quickly as possible. However, the objective of selective coordination is to disconnect a minimum amount of equipment from the power system. Coordination is obtained by selecting the appropriate type of circuit breaker, trip characteristics, and trip settings so that only the circuit breaker closest to the overload or short-circuit condition clears the problem.

To obtain selective coordination over the entire range of available short-circuit current, delayed operation of the line-side circuit breaker is necessary to allow the load-side circuit breaker to clear a fault. The additional time delay of the line-side device can increase the extent of the damage to circuit components. However, it is often necessary to make a compromise in protection to obtain coordination, within the limits of the National Electric Code (NFPA 70-2005) and ANSI protection requirements. Coordination between devices is often sacrificed when continuity of service and/or equipment damage and associated cost are not critical.

Chapter 6: Fused and special-purpose circuit breakers

This chapter discusses application of the following special-purpose circuit breakers:

- Instantaneous trip only
- Mine duty
- Current limiting
- Molded-case switch
- Integrally fused
- Circuit breaker and ground fault circuit interrupters
- Arc-fault circuit interrupters
- Supplementary protectors

Chapter 7: Acceptance and maintenance requirements.

This chapter is divided into the following two time periods:

- Acceptance testing that should be performed *before* placing a circuit breaker into service
- Maintenance guidelines that should be followed *after* a circuit breaker is placed into service

Acceptance testing and maintenance are required to ensure that the circuit breaker will perform satisfactorily to the full extent of the manufacturer's ratings. In addition, proper maintenance is required for continued, reliable operation. Also discussed briefly are testing and maintenance for insulated case circuit breakers, molded-case switches, and ground fault circuit interrupters. Annex 7A and Annex 7B provide sample data record sheets for recording circuit breaker test information and results.

A circuit breaker may be damaged during shipment, or defective components may be present; therefore, acceptance testing should be performed prior to placing it in service. Acceptance testing of circuit breakers is the initial testing on a breaker before it is placed into service to verify that the circuit breaker will perform properly. This chapter provides instructions for performing and documenting test results. Some acceptance test criteria discussed are insulation resistance, electrical operations at selected settings, mechanical operation, and auxiliary functions. Acceptance test data sheets are provided.

Maintenance procedures are important to power system reliability and safety. Proper maintenance will improve confidence that the circuit breakers will operate properly when called on to interrupt an overload or fault condition. The operating performance of a circuit breaker may deteriorate with time due to contamination by pollutants in the atmosphere, excessive use, or lack of operation for a prolonged duration. Welding in the main contact may occur during a short circuit if the contacts are in a deteriorated condition due to excessive use. In addition, the performance characteristics of other components in the breaker, such as contact pressure springs, may be degraded for some other reason such as overheating. This chapter discusses proper maintenance procedures and schedules that should be followed to maintain reliable operation.

1.6 Summary

This recommended practice identifies the differences between the two classifications of low-voltage circuit breakers and provides information for selecting the best one for a particular situation. In addition, this recommended practice gives information for applying circuit breakers at different locations in the power system and for protecting specific components. Guidelines are also given for coordinating combinations of line-side and load-side devices. Acceptance testing and maintenance guidelines are given so that reliable operation can be verified and maintained.

1.7 Normative references

The following referenced documents are indispensable for the application of this recommended practice. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

ANSI C37.16-2000, American National Standard Low-Voltage Power Circuit Breakers and AC Power Circuit Protectors-Preferred Ratings, Related Requirements, and Application Recommendations.³

ANSI C37.17-1997, American National Standard for Trip Devices for AC and General Purpose DC Low Voltage Power Circuit Breakers.

³ANSI publications are available from the Sales Department, American National Standards Institute, 25 West 43rd Street, 4th Floor, New York, NY 10036, USA (<http://www.ansi.org/>).

ANSI C37.50-1989 (R2000), American National Standard for Switchgear-Low-Voltage AC Power Circuit Breakers Used in Enclosures-Test Procedures.

IEEE Std C37.13-1990 (R1995), IEEE Standard for Low-Voltage AC Power Circuit Breakers Used in Enclosures.^{4, 5}

IEEE Std C37.100-1992, (R2001) IEEE Standard Definitions for Power Switchgear.

NFPA 70-2005, National Electrical Code[®] (NEC[®]).⁶

UL 489-2002, Molded-Case Circuit Breakers, Molded-Case Switches and Circuit-Breaker Enclosures.⁷

UL 1558-1999, Metal-Enclosed Low-Voltage Power Circuit Breaker Switchgear.

1.8 Bibliography

[B1] IEEE 100, *The Authoritative Dictionary of IEEE Standards Terms*, Seventh Edition. New York: Institute of Electrical and Electronics Engineers, Inc.

[B2] IEEE Std 141[™]-1993, IEEE Recommended Practice for Electric Power Distribution for Industrial Plants (*IEEE Red Book[™]*).

[B3] IEEE Std 242[™]-2001, IEEE Recommended Practice for Protection and Coordination of Industrial and Commercial Power Systems (*IEEE Buff Book[™]*).

⁴ IEEE publications are available from the Institute of Electrical and Electronics Engineers, 445 Hoes Lane, Piscataway, NJ 08855-1331, USA (<http://standards.ieee.org/>).

⁵The IEEE standards or products referred to in this clause are trademarks of the Institute of Electrical and Electronics Engineers, Inc.

⁶ NFPA publications are available from Publications Sales, National Fire Protection Association, 1 Batterymarch Park, P.O. Box 9101, Quincy, MA 02269-9101, USA (<http://www.nfpa.org/>).

⁷UL standards are available from Global Engineering Documents, 15 Inverness Way East, Englewood, Colorado 80112, USA (<http://global.ihs.com/>).

Chapter 2

Definitions, acronyms, and abbreviations

2.1 Definitions

For the purposes of this document, the following terms and definitions apply. *The Authoritative Dictionary of IEEE Standards Terms* [B1]¹ should be referenced for terms not defined in this clause.

2.1.1 adjustable: (as applied to circuit breakers) A qualifying term indicating that the circuit breaker can be set to trip at various values of current, time, or both, within a predetermined range. [adapted from the National Electrical Code[®] (NEC[®]) (NFPA 70-2005)]²

2.1.2 alarm switch: An auxiliary switch that actuates a signaling device upon the automatic opening of the circuit breaker with which it is associated.

2.1.3 auxiliary switch: A switch that is mechanically operated by the main switching device for signaling, interlocking, or other purposes.

NOTE—Auxiliary switch contacts are classified as a, b, aa, bb, LC, etc., for the purpose of specifying definite contact positions with respect to the main device.³

2.1.4 available short-circuit current: (at a given point in a circuit) The maximum current that the power system can deliver through a given circuit to any negligible-impedance short circuit applied at the given point, or at any other point that will cause the highest current to flow through the given point. *See also:* **prospective fault current.**

2.1.5 circuit breaker: A device designed to open and close a circuit by nonautomatic means, and to open the circuit automatically on a predetermined overcurrent without damage to itself when properly applied within its rating. [adapted from the NEC] *Syn:* **low-voltage circuit breaker.**

2.1.6 continuous load: A load where the maximum current is expected to continue for 3 hours or more. [adapted from the NEC]

2.1.7 coordination: The selection and/or setting of protective devices to provide protection of system components and to provide selectivity when a short circuit or other abnormality occurs.

2.1.8 current-limiting circuit breaker: A circuit breaker that does not use a fusible element and that when operating within its current-limiting range limits the let-through I^2t

¹The numbers in brackets correspond to those of the bibliography in 2.4.

²Information on references can be found in 2.3.

³Notes in text, tables, and figures are given for information only and do not contain requirements needed to implement the standard.

to a value less than the I^2t of a 1/2 cycle wave of the symmetrical prospective current. (adapted from UL 489-2002) *See also:* **current-limiting overcurrent protective device.**

2.1.9 current-limiting overcurrent protective device: A device that, when interrupting currents in its current-limiting range, reduces the current flowing in the faulted circuit to a magnitude substantially less than that obtainable in the same circuit if the device were replaced with a solid conductor having comparable impedance. (adapted from the NEC) *See also:* **current-limiting circuit breaker.**

2.1.10 drawout-mounted circuit breaker: An assembly of a circuit breaker together with a supporting structure so constructed that the breaker is supported and can be moved between the main circuit connected and the disconnected positions without unbolting connections or mounting supports. The stationary portion of the drawout assembly includes self-supporting primary circuit terminals and may include an interlocking means that permits movement of the breaker between the main circuit connected and the disconnected positions only when the breaker contacts are in the open position. (adapted from UL 489-2002)

2.1.11 dynamic impedance: The arc impedance introduced into a circuit by the opening of the circuit-breaker contacts during current interruption. (adapted from NEMA AB 1-1993)

2.1.12 electrical operator: A controlling device that is used to open, close, and reset a circuit breaker. (adapted from UL 489-2002)

NOTE—The term *motor operator* is sometimes used when the operating device is a motor.

2.1.13 electronic trip unit: A self-contained portion of a circuit breaker that senses the condition of the circuit breaker electronically and that actuates the mechanism that opens the circuit breaker contacts automatically.

NOTE—Where the term *electronic trip unit* is used in this book, the alternate terms *solid state trip unit* and *static trip unit* are commonly used in literature.

2.1.14 frame: As applied to circuit breakers, that portion of an interchangeable trip unit circuit breaker remaining when the interchangeable trip unit is removed. (adapted from UL 489-2002)

2.1.15 frame size: A term applied to a group of circuit breakers of similar physical configuration. Frame size is expressed in amperes and corresponds to the largest ampere rating available in the group. The same frame size designation may be applied to more than one group of circuit breakers. (adapted from UL 489-2002)

2.1.16 ground-fault delay: An intentional time delay in the tripping of a circuit breaker when a ground fault occurs. (adapted from NEMA AB 1-1993)

2.1.17 ground-fault pickup: The nominal value of the ground fault current at which the ground fault delay function is initiated. (adapted from UL 489-2002)

2.1.18 ground-fault protection of equipment: A system intended to provide protection of equipment from damaging line-to-ground fault currents by operating to cause a disconnecting means to open all ungrounded conductors of the faulted circuit. This protection is provided at current levels less than those required to protect conductors from damage through the operation of a supply circuit overcurrent device. (adapted from the NEC)

2.1.19 I^2t : An expression related to the energy available as a result of current flow, meaningful only for adiabatic conditions. With respect to circuit breakers, the expression refers to the I^2t between the initiation of fault current and the clearing of the circuit. The defining equation is $I^2t = \int I^2(t) dt$ over the stated period, in units of amperes-squared seconds. (adapted from UL 489-2002)

2.1.20 instantaneous pickup: The nominal value of current at which an adjustable circuit breaker is set to trip instantaneously. (adapted from UL 489-2002)

2.1.21 instantaneous trip: (as applied to circuit breakers) A qualifying term indicating that no delay is purposely introduced in the tripping action of the circuit breaker. (adapted from the NEC)

2.1.22 instantaneous-trip-only circuit breaker: A circuit breaker intended to provide short-circuit protection only. Although acting instantaneously under short-circuit conditions, instantaneous trip breakers shall be permitted to include a transient damping action to ride through initial motor transients. (adapted from UL 489-2002)

2.1.23 insulated-case circuit breaker (ICCB): A circuit breaker that is assembled as an integral unit in a supporting and enclosing housing of insulating material and with a stored energy mechanism.

2.1.24 integrally-fused circuit breaker: A circuit breaker in which coordinated fuses are connected in series with the release (trip) elements of the circuit breaker and are mounted within the housing of the circuit breaker. (adapted from NEMA AB 1-1993)

2.1.25 interrupting rating: The highest current at rated voltage that a device is intended to interrupt under standard test conditions. (adapted from the NEC)

2.1.26 inverse time: (as applied to circuit breakers) A qualifying term indicating that there is purposely introduced a delay in the tripping action of the circuit breaker, which delay decreases as the magnitude of the current increases. (adapted from the NEC)

2.1.27 I_p : *See:* **peak current.**

2.1.28 long-time delay: An intentional time delay in the overload tripping of an adjustable circuit breakers inverse time characteristic. (adapted from UL 489-2002)

2.1.29 long-time pickup: The current at which the long-time delay function is initiated. (adapted from UL 489-2002)

2.1.30 low-voltage power circuit breaker (LVPCB): A mechanical switching device, capable of making, carrying, and breaking currents under normal circuit conditions and making and carrying for a specified time and breaking currents under specified abnormal circuit conditions such as those of short circuit. Rated 1000 V ac or below, or 300 V dc and below, but not including molded-case circuit breakers. (adapted from IEEE Std C37.100™-1992)

2.1.31 molded-case circuit breaker (MCCB): A circuit breaker that is assembled as an integral unit in a supporting and enclosing housing of insulating material. (adapted from UL 489-2002 and IEEE Std C37.100-1992)

2.1.32 molded-case switch: A device designed to open and close a circuit by nonautomatic means, assembled as an integral unit in a supportive and enclosed housing of insulating material. *Syn:* **nonautomatic switch** (deprecated). (adapted from UL 489-2002 and UL 1087-1993)

2.1.33 overcurrent: Any current in excess of the rated current of equipment or the ampacity of a conductor. It may result from overload, short circuit, or ground fault. (adapted from the NEC) *See also:* **overload**.

2.1.34 overload: Operation of equipment in excess of normal, full-load rating or of a conductor in excess of rated ampacity that, when it persists for a sufficient length of time, would cause damage or dangerous overheating. A fault, such as a short circuit or ground fault, is not an overload. (adapted from the NEC) *See also:* **overcurrent**.

2.1.35 panelboard: A single panel or group of panel units designed for assembly in the form of a single panel, including buses and automatic overcurrent devices, and equipped with or without switches for the control of light, heat, or power circuits; designed to be placed in a cabinet or cutout box placed in or against a wall, partition, or other support; and accessible only from the front. (adapted from the NEC) *See also:* **switchboard**.

2.1.36 peak current: The maximum instantaneous current that flows in a circuit, designated I_p . (adapted from UL 489-2002)

2.1.37 peak let-through current: The highest current flowing in the circuit after the inception of the fault that the circuit breaker and the protected system must withstand, expressed as an instantaneous rather than a root-mean-square (rms) value.

2.1.38 pickup: The root-mean-square (rms) current at which a circuit breaker tripping function is initiated. (adapted from NEMA AB 1-1993)

2.1.39 prospective fault current: The current that would flow during a short circuit if the circuit breaker and the wires used for its connection were replaced by a solid conductor of negligible impedance. (adapted from UL 489-2002) *See also:* **available short-circuit current**.

2.1.40 rated short-time withstand current: (A) The maximum root-mean-square (rms) total current that a circuit breaker can carry momentarily without electrical, thermal, or

mechanical damage or permanent deformation. The current shall be the rms value, including the dc component, at the major peak of the maximum cycle as determined from the envelope of the current wave during a given test time interval. (adapted from IEEE Std C37.100-1992) **(B)** That value of current assigned by the manufacturer that the device can carry without damage to itself, under prescribed conditions. (adapted from NEMA AB 1-1993) *Syn:* **withstand rating; short-time rating.**

2.1.41 rating plug: An interchangeable module of an electronic trip unit that, together with the sensor, sets the current rating range of the circuit breaker. For example, a 1200 A frame may contain an 800 A sensor, fixing the maximum rating that can be configured for the unit at 800 A adjustable by the following kind of settings. By installing a 600 A rating plug, the adjustable rating is correspondingly 600 A multiplied by the long-time pickup adjustment [i.e., the long-time pickup may be adjusted to “0.9,” and the ampere rating or setting is $(0.9 \times 600 \text{ A}) = 540 \text{ A}$]

2.1.42 release: A device, mechanically connected within the circuit breaker, that initiates the tripping function of a circuit-breaker. (adapted from NEMA AB 1-1993)

2.1.43 root-mean-square (rms) sensing: A term commonly used to indicate the sensing of root-mean-square (rms) value current rather than instantaneous or peak values, as by a circuit-breaker trip unit.

2.1.44 selective coordination: *See:* **coordination; selectivity.**

2.1.45 selectivity: A general term describing the interrelated performance of relays and breakers and other protective devices; complete selectivity being obtained when a minimum amount of equipment is removed from service for isolation of a fault or other abnormality. (adapted from IEEE Std C37.100-1992) *See also:* **coordination.**

2.1.46 sensor: (as applied to a circuit-breaker with an electronic trip unit) A current sensing element such as a current transformer within a circuit-breaker frame. The sensor will have a current rating less than or equal to the frame size and will provide the sensing function for a specific group of current ratings within the frame size.

2.1.47 series rating: The interrupting rating of a tested combination of a line-side (main) overcurrent protective device and a load-side (branch) circuit-breaker in which the interrupting rating of the combination is greater than the interrupting rating of the branch circuit-breaker. The interrupting rating of the series combination does not exceed the interrupting rating of the main overcurrent protective device. *Syn:* **series-connected rating.**

NOTE—See *UL Recognized Component Directory*, 2002 [B2].

2.1.48 setting: (of a circuit breaker) The value of current, time, or both at which an adjustable circuit-breaker is set to trip. (adapted from the NEC)

2.1.49 short circuit: An abnormal connection (including an arc) of relatively low impedance, whether made accidentally or intentionally, between two points of different potential. (adapted from IEEE Std C37.100-1992)

2.1.50 short-time current: The current carried by a device, an assembly, or a bus for a specified short-time interval. (adapted from IEEE Std C37.100-1992) *See also:* **rated short-time withstand current; short-time rating.**

2.1.51 short-time delay: An intentional time delay in the tripping of a circuit-breaker that is above the overload pickup setting.

2.1.52 short-time delay phase or ground trip element: A direct-acting trip device element that functions with a purposely delayed action (measured in milliseconds). (adapted from IEEE Std C37.100-1992)

2.1.53 short-time pickup: The current at which the short-time delay function is initiated. (adapted from UL 489- 2002)

2.1.54 short-time rating: A rating applied to a circuit breaker that, for reason of system coordination, causes tripping of the circuit breaker to be delayed beyond the time when tripping would be caused by an instantaneous element. (adapted from UL 489-2002) *See also:* **rated short-time withstand current; short-time current.**

2.1.55 shunt trip device: A trip mechanism energized by a source of voltage that may be derived either from the main circuit or from an independent source. (adapted from UL 489-2002)

2.1.56 stored-energy operation: Operation by means of energy stored in the mechanism before the completion of the operation and sufficient to complete it under predetermined conditions. (adapted from IEEE Std C37.100-1992)

2.1.57 switchboard: A large, single-panel, frame or assembly of panels on which are mounted, on the face, back, or both, switches, overcurrent and other protective devices, buses, and usually instruments. Switchboards are generally accessible from the rear as well as from the front and are not intended to be installed in cabinets. (adapted from the NEC) *See also:* **panelboard.**

2.1.58 switchgear: A general term covering switching and interrupting devices and their combination with associated control, metering, protective and regulating devices; also, assemblies of these devices with associated interconnections, accessories, enclosures, and supporting structures used primarily in connection with the generation, transmission, distribution, and conversion of electric power. (adapted from IEEE Std C37.100-1992)

2.1.59 transient recovery voltage (TRV): The voltage transient that occurs across the terminals of a pole of a switching device upon interruption of the current. (adapted from IEEE Std C37.100-1992)

2.1.60 tripping: The opening of a circuit breaker by actuation of the release mechanism. (adapted from UL 489- 2002)

2.1.61 trip unit: A self-contained portion of a circuit breaker that actuates the mechanism that opens the circuit-breaker contacts automatically. (adapted from UL 489- 2002)

NOTE—The terms *trip device* and *tripping device* are used in literature as alternative terms for trip unit.

2.1.62 undervoltage trip device: A trip mechanism that causes a circuit-breaker to open automatically if the voltage across the terminals of the trip coil falls below a predetermined value. (adapted from UL 489- 2002) *Syn:* **undervoltage release.**

2.1.63 withstand current: *See:* **rated short-time withstand current.**

2.1.64 withstand rating: *See:* **rated short-time withstand current.**

2.1.65 zone selective interlocking: A function provided for rapid clearing while retaining coordination. The function is a communication interconnection between the electronic trip units of two or more circuit breakers connected in series on multiple levels. By means of intercommunication between the short-time delay and/or ground fault elements, the one nearest the fault trips with minimum time delay while signaling the supply-side circuit-breaker(s) to delay for a predetermined period. *Syn:* **zone interlocking; selective interlocking.**

2.2 Acronyms and abbreviations

2.2.1 40 °C: Designates a circuit-breaker that is acceptable for use in ambient temperatures up to 40 °C.

2.2.2 AIC: Amperes interrupting capacity. Maximum current a protective device is capable of interrupting, namely, its interrupting rating. May be found in manufacturer's literature. Also see **AIR**.

2.2.3 AIR: Amperes interrupting rating. Shortened term marked on some small circuit breakers with the interrupting rating.

2.2.4 CTL: A Class CTL circuit breaker, because of its size or configuration in conjunction with the physical means provided in Class CTL panelboards, prevents more circuit-breaker poles from being installed than the number for which the assembly is designed and rated. A Class CTL panelboard is a circuit-limited lighting and appliance panelboard as referenced in Sections 408.14 to 408.16 of the NEC. Both half-size and full-sized circuit breakers may be installed.

2.2.5 HACR: Heating, air conditioning, and refrigeration. Designates compliance with the special requirements of Section 430-53(C)(3) of the NEC, as listed for group installation for use with heating, air conditioning, and refrigeration equipment. A circuit breaker with

this marking is suitable only for use with equipment marked to indicate that an HACR circuit breaker is acceptable.

2.2.6 HID: High-intensity discharge. Indicates construction suitable for switching high-intensity discharge lighting loads.

2.2.7 SWD: Switching duty. Designates compliance with requirements for circuit breakers used as switches on fluorescent lighting circuits as indicated in Section 240-83(D) of the NEC.

2.3 Normative references

The following referenced documents are indispensable for the application of this standard. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

IEEE Std C37.100-1992 (R2001), IEEE Standard Definitions for Power Switchgear.⁴

NEMA AB 1-1993, Molded Case Circuit Breakers and Molded Case Switches.⁵

NFPA 70-2005, National Electrical Code[®] (NEC[®]).⁶

UL 489-2002, Molded-Case Circuit Breakers, Molded-Case Switches, and Circuit-Breaker Enclosures.⁷

UL 1087-1993, Molded-Case Switches.

UL Molded-Case Circuit Breaker Marking Guide, 1996.

2.4 Bibliography

[B1] IEEE 100, *The Authoritative Dictionary of IEEE Standards Terms*, Seventh Edition. New York: Institute of Electrical and Electronics Engineers, Inc.

[B2] *UL Recognized Component Directory*, 2002.

⁴IEEE publications are available from the Institute of Electrical and Electronics Engineers, 445 Hoes Lane, Piscataway, NJ 08855-1331, USA (<http://standards.ieee.org/>).

⁵NEMA publications are available from Global Engineering Documents, 15 Inverness Way East, Englewood, CO 80112, USA (<http://global.ihs.com/>).

⁶NFPA publications are available from Publications Sales, National Fire Protection Association, 1 Batterymarch Park, P.O. Box 9101, Quincy, MA 02269-9101, USA (<http://www.nfpa.org/>).

⁷UL standards are available from Global Engineering Documents, 15 Inverness Way East, Englewood, Colorado 80112, USA (<http://global.ihs.com/>).